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Domination Parameters of Graph Covers

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Abstract

A graph G is a *cover* of a graph F if there exists an onto mapping $\pi : V(G) \to V(F)$, called a (*covering*) *projection*, such that π maps the neighbours of any vertex v in G bijectively onto the neighbours of $\pi(v)$ in F. This paper is the first attempt to study the connection between domination parameters and graph covers. We focus on the domination number, the total domination number, and the connected domination number. We prove upper and lower bounds for the domination parameters of G.

Keywords: domination, total domination, connected domination, graph cover.

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1 Introduction

In this paper, we deal with finite undirected and simple graphs. For a graph F, let $V(F)$ and $E(F)$ respectively denote the vertex set and the edge set of F.

A graph G is a *cover* of a graph F if there exists an onto mapping $\pi : V(G) \to V(F)$, called a (*covering*) *projection*, such that π maps the neighbours of any vertex v in G bijectively onto the neighbours of $\pi(v)$ in F. Note that every graph is a cover of itself via the identity projection. If F is connected, then $|\pi^{-1}(v)| = k$ is the same for all $v \in V(F)$ and π is called a *k-fold* cover. We often call a 2-fold cover a *double cover*. See Figure 1 for an example of a cover. Here, we can define the vertex projection π by labelling every vertex v of G with their image $\pi(v)$. We often add a dash or subscript to distinguish between two or more vertices of G projected to the same vertex of F. Let graph G be a

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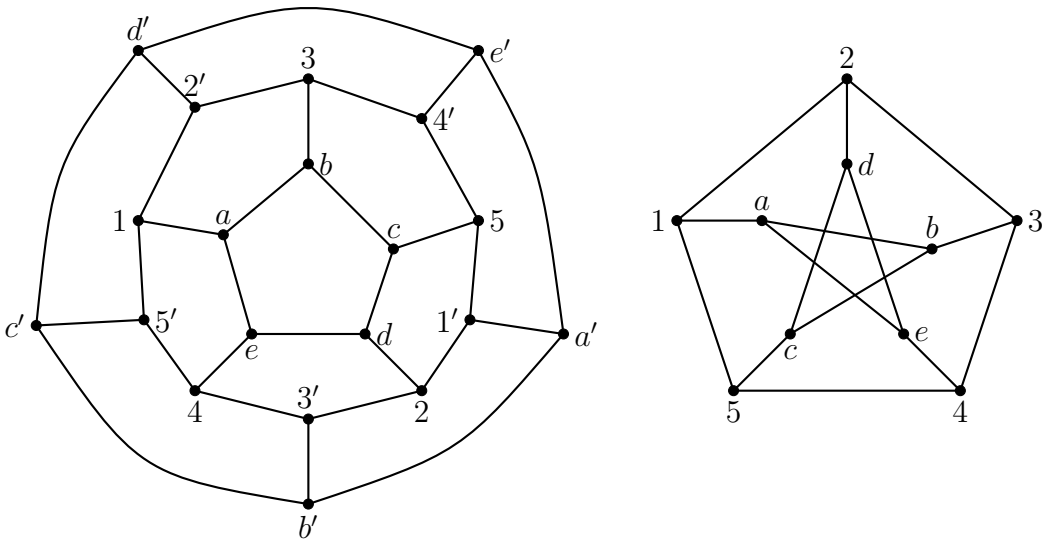


Figure 1: Dodecahedron (left) is a (double) cover of the Petersen graph (right).

cover of graph F. For any subgraph F' of F, we call the graph $G' = \pi^{-1}(F')$ the *lift of* F' *into* G.

The following proposition illustrates some properties of covers.

Proposition 1. *Let graph G be a cover of graph F via the projection π.*

(i) It holds that $\deg_G(v) = \deg_F(\pi(v))$ *for each vertex* $v \in V(G)$*, where* $\deg_G(v)$ *is the degree of a vertex v in G.*

(ii) The lift of a tree T into G consists of a collection of disjoint trees isomorphic to T. Hence, a cover of a tree is a forest.

(iii) The lift of a cycle C_n into G consists of a collection of disjoint cycles whose lengths are multiples of n.

A lot of attention has been paid to graph covers from various perspectives. As purely graph-theoretic objects, graph covers were studied in [4, 15, 19]. Graph covers have also appeared in the literature as "lifts" of graphs, see [2].

Let S be a subset of the vertex set of graph F. S is called a *dominating set* of F if every vertex not in S is adjacent to at least one vertex in S. The *domination number* of F, denoted by $\gamma(F)$, is the minimum cardinality of a dominating set of F. A *total dominating set* of F is a set S such that every vertex is adjacent to some vertex in S. Note that by the definition of a total dominating set, it is evident that a graph F admits a total dominating set if and only if F has no isolated vertices. The *total domination number* of F, denoted by $\gamma_t(F)$, is the minimum cardinality of a total dominating set

of F. A *connected dominating set* S of F is a set such that every vertex not in S is adjacent with at least one vertex in S and the subgraph induced by S is connected. The *connected domination number* $\gamma_c(F)$ is the minimum cardinality of a connected dominating set of F.

We saw in Figure 1 that the dodecahedron is a cover of the Petersen graph. One can check that $\gamma(\text{Petersen graph}) = 3$ and $\gamma_t(\text{Petersen graph}) = \gamma_c(\text{Petersen graph}) = 4$. Moreover, $\gamma(\text{Dodecahedron}) = 6, \gamma_t(\text{Dodecahedron}) = 8$ and $\gamma_c(\text{Dodecahedron}) = 10$.

The theory of domination in graphs is one of the main research areas in graph theory. For an excellent treatment of fundamentals of domination in graphs and recent topics, we refer the reader to the following books [16–18].

One fundamental area of study in domination in graphs is how domination parameters change under graph operations. For example, Vizing [25] studied the connection between the domination number and the cartesian product of graphs, which led to Vizing's conjecture, which many believe to be the main open problem in the area of domination in graphs. Also, see [6, 11, 14, 22, 27] for other similar research. Graph covers give an interesting way to construct new graphs that are locally isomorphic to a given graph. As a result, graph covers have a lot of intriguing relationships with different types of graph parameters. For example, connections between connectivity and graph covers were carried out in [1, 3, 20]. Also, see [2] for other interesting graph-theoretic connections.

The preceding paragraph serves as the main motivation for this work. Let graph G be a cover of graph F. Our goal is to study the relation between $\gamma(G)$ and $\gamma(F)$, $\gamma_t(G)$ and $\gamma_t(F)$, and $\gamma_c(G)$ and $\gamma_c(F)$. To our knowledge, this is the first paper to discuss the connections between domination parameters and graph covers.

Now we present theorems that are useful in our results. Note that from Proposition 1(i) we have $\Delta(F) = \Delta(G)$ and $\delta(G) = \delta(F)$. So, we write Δ and δ to mean the maximum and minimum degree, respectively, without specifying the graph.

Theorem 2 ([26]). *If F is a graph of order n, then $\gamma(F) \geqslant \frac{n}{1+\Delta}$.*

Theorem 3 ([21]). *If F is a graph of order n with $\delta \geqslant 3$, then $\gamma(F) \leqslant \frac{3}{8}n$.*

Theorem 4 ([7]). *If F is a graph of order n with $\delta \geqslant 5$, then $\gamma(F) \leqslant \frac{1}{3}n$.*

Theorem 5 ([9, 26]). *If F is a connected graph of order n, then $\gamma_t(F) \geqslant \frac{n}{\Delta}$.*

Theorem 6 ([5, 8, 23]). *If F is a graph of order n with $\delta \geqslant 3$, then $\gamma_t(F) \leqslant \frac{1}{2}n$.*

Theorem 7 ([10]). *If F is a graph of order n with $\delta \geqslant 5$, then $\gamma_t(F) \leqslant \frac{2453}{6500}n$.*

Greedy Algorithm

There is an intuitive algorithm which constructs a dominating subset of vertices of a graph F whose cardinality differs from $\gamma(F)$ by no more than, roughly, $\log \Delta$.

Given a graph F. Let S be a subset of the vertex set of F. We start with $S = \varnothing$. Until S is a dominating set of F, add a vertex v with maximum number of undominated neighbours. All vertices of F start as *white*. Anytime we add a vertex to S, we colour that vertex *black*, and colour all dominated vertices *grey*. Note that all undominated vertices remain white. Let $w(u)$ be the cardinality of the white vertices among the neighbours of u, including u itself. This idea leads to a greedy algorithm.

Algorithm 1 Greedy algorithm

1. $S := \varnothing$
2. **while** $\exists$ white vertices **do**
3. choose $v \in \{x \mid w(x) = \max_{u \in V}\{(w(u)\}\}$;
4. $S := S \cup \{v\}$;
5. **end while**

Theorem 8 ([24])**.** *The greedy algorithm (see Algorithm 1) is a $H(\Delta)$ factor approximation algorithm for the minimum dominating set problem, where $H(\Delta) = 1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{\Delta}$.*

2 Results

This section is devoted to establishing relations concerning the domination number, the total domination number, and the connected domination number with respect to graph covers. We start with the domination number.

Theorem 9. *Suppose that a graph G is a k-fold cover of a graph F. Then $\gamma(G) \leqslant k\gamma(F)$. Moreover, this bound is tight.*

Proof. Suppose that $\gamma(F) = t$. Let π be the projection map, and let $S = \{v_1, v_2, \ldots, v_t\}$ be a minimal dominating set of F such that $|S| = t$. We define the set $S' = \{\pi^{-1}(v) \mid v \in S\}$. We show that S' is a dominating set of G. It is sufficient to show that $v' \in V(G) \setminus S'$ has at least one neighbour in S'. Suppose for a contradiction that there is $v' \in V(G) \setminus S'$ such that v' has no neighbour in S'. Then $\pi(v')$ has no neighbour in S, but S was chosen to be a dominating set with the smallest cardinality; a contradiction. Hence v' has a neighbour in S' and furthermore, S' is a dominating set. It follows that $\gamma(G) \leqslant |S'| = kt$.

The bound is trivially tight because if F is any graph and G is k disconnected copies of F, then $\gamma(G) = k\gamma(F)$. □

A natural question that arises is: can we find a lower bound for $\gamma(G)$ in terms of k and $\gamma(F)$?

We make the following observations.

Observation 1.

(i) If a graph G is a k-fold cover of a graph F, then $\gamma(G) \geqslant k$ by Theorem 2.
(ii) If a graph G is a k-fold cover of a graph F, then $\gamma(G) \geqslant \gamma(F)$.

The following is an immediate consequence of Observation 1.

Remark 1. Suppose that a graph G is a k-fold cover of a graph F. Then $\gamma(G) \geqslant \sqrt{k\gamma(F)}$.

The bound in Remark 1 is not strong. In the following, we shall prove a stronger lower bound, which depends on Δ.

Theorem 10. *Suppose that a graph G is a k-fold cover of a graph F. Then $\gamma(G) \geqslant \frac{1}{H(\Delta)}k\gamma(F)$, where $H(\Delta) = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{\Delta}$.*

Proof. Let $S = \{v_1, v_2, \ldots, v_\ell\}$ be a set of vertices in a dominating set of F chosen by the Greedy algorithm, where v_i was chosen before v_{i+1}, for $i = 1, 2, \cdots, \ell - 1$. Then $\gamma(F) \leqslant |S|$, so $k\gamma(F) \leqslant k|S| = |\pi^{-1}(S)|$, (where π is the projection map for the cover).

We will show that the lift of S into G could be the set chosen by the greedy algorithm. Let $S' = \varnothing$ be a subset of the vertex set of G. Note that all the vertices of G are white. Since $v_1 \in S$ was chosen first among all the vertices of F, the greedy algorithm could choose vertices of G that are in $\pi^{-1}(v_1)$ because each vertex in $\pi^{-1}(v_1)$ together with their neighbours form a collection of disjoint stars in G. Since there are k vertices in $\pi^{-1}(v_1)$, the greedy algorithm could add all the k preimages $\pi^{-1}(v_1)$ of v_1 to S' . So, $S' := S' \cup \{v'_{11}, v'_{12}, \ldots, v'_{1k}\}$, where v'_{1i}, for $i = 1, 2, \ldots, k$, are vertices in $\pi^{-1}(v_1)$. Note that π maps the neighbours of v'_{1i}, (for $i = 1, 2, \ldots, k$) bijectively onto the neighbours of v_1. So, $w(u)$ maintains the same value as $w(\pi^{-1}(u))$, for all $u \in S \setminus v_1$.

Now, the next vertex in S is v_2. Again, the greedy algorithm could choose vertices of G that are in $\pi^{-1}(v_2)$. Since there are k preimages $\pi^{-1}(v_2)$ of v_2, S' becomes $S' = S' \cup \{v'_{11}, v'_{12}, \ldots, v'_{1k}\} \cup \{v'_{21}, v'_{22}, \ldots, v'_{2k}\}$, where v'_{1i} and $v'_{2,i}$ are vertices in $\pi^{-1}(v_1)$ and $\pi^{-1}(v_2)$, respectively, for $i = 1, 2, \ldots, k$. By repeating this process for the sequence

of vertices in S, it follows that the lift of S into G could be a chosen set by the greedy algorithm for G. Thus, from Theorem 8 we have $|\pi^{-1}(S)| \leqslant \gamma(G)H(\Delta)$ and so, $k\gamma(F) \leqslant \gamma(G)H(\Delta)$. This proves the theorem. □

The following result is a consequence of Theorem 10.

Corollary 11. *Let a graph G be a k-fold cover of C_n, a cycle on n vertices. Then $\gamma(G) \geqslant \frac{2}{3}k\gamma(C_n)$.*

Proof. Note that $H(2) = \frac{3}{2}$, and the result follows from Theorem 10. □

For r-regular graphs with $3 \leqslant r \leqslant 5$, we have the following lower bound.

Theorem 12. *Suppose that a graph G is a k-fold cover of a graph F. Then*

(a) $\gamma(G) \geqslant \frac{3}{5}k\gamma(F)$ *if F is a 3-regular graph.*

(b) $\gamma(G) \geqslant \frac{1}{2}k\gamma(F)$ *if F is a 4 or 5-regular graph.*

Proof. For assertion (a), suppose for a contradiction that $\gamma(G) < \frac{3}{5}k\gamma(F)$. Let F be a 3-regular graph on n vertices. Note that G has kn vertices. Then by applying Theorems 2 and 3 we have

$$\frac{1}{4}kn \leqslant \gamma(G) < \frac{3}{5}k\gamma(F) \leqslant \frac{9}{40}kn,$$

which leads to a contradiction.

For assertion (b), the proof is similar to the proof for assertion (a) by using Theorems 2, 3 and 4. □

For $r = 3, 4$ and 5, Theorem 10 gives the following leading coefficients $\frac{6}{11}, \frac{12}{25}$ and $\frac{60}{137}$, respectively. So, Theorem 12 gives better leading coefficients than Theorem 10 when $r = 3, 4, 5$. Moreover, we do not have examples to illustrate that the bounds in Theorems 10 and 12 are tight. Therefore, we think that the bound could be improved generally.

Example 13. In Figure 2, the graph N is the Cartesian product of C_3 and C_3, where the black vertices are dominating vertices. So $\gamma(N) \leqslant 3$. Note that N is 4-regular. So, we can choose any vertex to be in the dominating set and that vertex dominates four other vertices. Now, the remaining four undominated vertices lie on a rectangle in N. So we need at least two additional vertices to dominate all of them. Hence, $\gamma(N) = 3$.

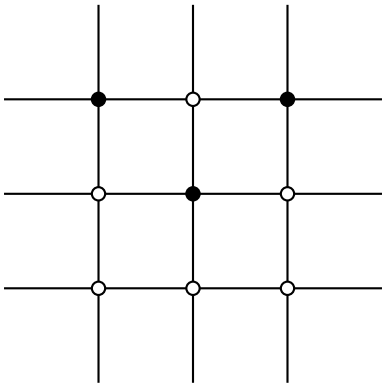


Figure 2: The graph N.

Let a graph G be the Cartesian product of C_{15} and C_{15}. Clearly, G is a cover of N with fold 25. We construct a dominating set S of G as follows. Choose any vertex, say v, of G as the starting point. Then v dominates four other vertices. We translate v both vertically and horizontally with a period of 5 [12]. We iterate this process for every new vertex we select. Since G is 4-regular and every vertex lies on C_{15} , we select three vertices on every cycle of length 15. So, every vertex in G is dominated by exactly one vertex of S, and therefore, S is a dominating set. Thus, $\gamma(G) = \frac{15^2}{5} = 45$ and moreover, $\gamma(G) = \frac{3}{5}k\gamma(N)$.

We now turn our attention to the total domination number. We prove a result analogous to Theorem 9.

Theorem 14. *Let F be a graph without isolated vertices. Suppose that G is a k-fold cover of F. Then $\gamma_t(G) \leqslant k\gamma_t(F)$. Also, this bound is tight.*

Proof. The proof of the inequality is similar to the proof for Theorem 9. The bound is trivially tight if we take G to be a disconnected union of k copies of F. □

We make the following observations, which will be used to prove a general lower bound for $\gamma_t(G)$.

Observation 2.

(i) If a graph G is a k-fold cover of graph F without isolated vertices, then $\gamma_t(G) \geqslant k$ from Theorem 5.

(ii) If a graph G is a k-fold cover of graph F without isolated vertices, then $\gamma_t(G) \geqslant \gamma_t(F)$.

Remark 2. Let F be a graph without isolated vertices. Suppose that a graph G is a k-fold cover of F. Then $\gamma_t(G) \geqslant \sqrt{k\gamma_t(F)}$.

For r-regular graphs with $3 \leqslant r \leqslant 5$, we have the following lower bound.

Theorem 15. *Suppose that a graph G is a k-fold cover of a graph F. Then*

(a) $\gamma_t(G) \geqslant \frac{3}{5}k\gamma_t(F)$ *if F is a 3-regular graph.*

(b) $\gamma_t(G) \geqslant \frac{1}{2}k\gamma_t(F)$ *if F is a 4 or 5-regular graph.*

Proof. Using Theorems 5, 6, and 7, one can prove the result in the same way as the proof for Theorem 12. □

Finally, we consider the connected domination number. Note that by definition, a graph admits a connected dominating set if and only if the graph is connected.

We observe the following.

Observation 3.

(i) If a connected graph G is a k-fold cover of a connected graph F, then $\gamma_c(G) \geqslant k$ because $\gamma_c(G) \geqslant \gamma(G)$.
(ii) If a connected graph G is a k-fold cover of a connected graph F, then $\gamma_c(G) \geqslant \gamma_c(F)$.

Remark 3. Let F and G be connected graphs. If G is a k-fold cover of F, then $\gamma_c(G) \geqslant \sqrt{k\gamma_c(F)}$.

Theorem 16. *Let F and G be connected graphs. Suppose that G is a k-fold cover of F. Then $\gamma_c(G) \leqslant k(\gamma_c(F) + 2) - 2$. Moreover, this bound is tight.*

Proof. Let S be a connected dominating set of F with a minimum cardinality. Then, by the definition, the elements of S induce a connected subgraph F' of F. Let T' be a spanning tree of F'. Then from Proposition 1 (ii), $\pi^{-1}(T')$ is a disjoint union of trees of G isomorphic to T'. Let $T_1, \ldots, T_k$ be the connected components of $\pi^{-1}(T')$. Note that the set of vertices of the subgraph $\cup_{i=1}^{k} T_i \subset G$ is a dominating set of G. We construct a connected subgraph G' of G containing the vertices of $\cup_{i=1}^{k} T_i$ and having no more than $k(\gamma_c(K) + 2) - 2$ vertices (note that $\cup_{i=1}^{k} T_i$ has $k\gamma_c(F)$ vertices). The subgraph G' is constructed by induction. We start with setting $G'_1 = T_1$ and with considering a shortest path p_1 between G'_1 and $\cup_{i=2}^{k} T_i$. Up to relabelling, we can assume that the endpoint of p_1 not belonging to G'_1 belongs to T_2. The path p_1 cannot have more than three edges,

as otherwise it would contain a vertex not adjacent to any vertices of $\cup_{i=1}^{k} T_i$. We define $G'_2 = G'_1 \cup p_1 \cup T_2$, and consider a shortest path p_2 between H_2 and $\cup_{i=3}^{k} T_i$. The path p_2 cannot have more than 3 edges. Up to relabelling we can assume that the endpoint of p_2 not belonging to G'_2 belongs to T_3, and we define $G'_3 = G'_2 \cup p_2 \cup T_3$. Repeating this procedure, we obtain a connected subgraph $G' = G'_{k-1} \subset G$ which contains $\cup_{i=1}^{k} T_i$ and whose number of vertices is at most $k\gamma_c(F) + 2(k-1)$, as required.

If $F = C_n$ (for $n \geqslant 3$), then $G = C_{kn}$. It is known that $\gamma_c(C_n) = n - 2$. So, $\gamma_c(G) = kn - 2 = k(n - 2 + 2) - 2$, which proves that the bound is tight. □

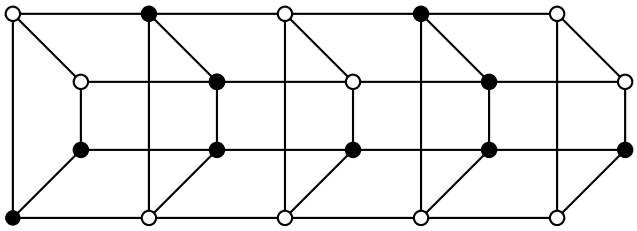


Figure 3: The graph H.

Example 17. In Figure 3, the black vertices are connected dominating set of a graph H. So, $\gamma_c(H) \leqslant 9$. Without loss of generality, we consider the top horizontal line in Figure 3. We can dominate the five vertices with exactly two vertices, and the two vertices cannot be adjacent. Since we are forming a connected dominating set, each of the two vertices must be adjacent to a vertex in the connected dominating set. So we select additional two vertices on the second-top horizontal line. Again, we can dominate the third-top horizontal line with two vertices and they are not adjacent. So, we have to select their common adjacent vertex to be in the connected dominating set. Finally, we dominate the bottom horizontal line with two vertices. Thus, $\gamma_c(H) = 9$. Let a graph G be the 5×8 grid with an edge connecting every pair u_i, v_i, for $i = 1, 2, \ldots, 5$, where u_i, v_i (for $i = 1, 2, \ldots, 5$) are the vertices of the opposite ends of the 5×8 grid, and u_i, v_i lie on the same straight line in the grid. Then G is a 2-fold cover of graph H. The connected domination number of the 5×8 grid is 17 (see [13]) and hence $\gamma_c(G) \leqslant 17$. But $17 < 2 \times 9 = 18$. So in general, it is not true that $\gamma_c(G) \geqslant k\gamma_c(F)$ if G is a k-fold cover of F, where G and F are connected graphs.

3 Conclusion

In this paper, we investigated how three domination parameters behave under graph covers. We hope that another major direction of research on domination in graphs

is suggested by the results. Furthermore, our results suggest that the problem of characterising graphs that achieve equality in any of the upper bounds would be interesting.

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References

[1] Alon Amit and Nathan Linial. Random graph coverings i: General theory and graph connectivity. *Combinatorica*, 22(1):1–18, 2002.

[2] Alon Amit, Nathan Linial, Jiří Matoušek, and Eyal Rozenman. Random lifts of graphs. In *Proceedings of the twelfth annual ACM-SIAM symposium on Discrete algorithms*, pages 883–894. Citeseer, 2001.

[3] Dickson Y. B. Annor, Yuri Nikolayevsky, and Michael S. Payne. Three theorems on Negami's planar cover conjecture. *arXiv preprint arXiv:2412.19560*, 2024.

[4] Dan Archdeacon and R Bruce Richter. On the parity of planar covers. *Journal of graph theory*, 14(2):199–204, 1990.

[5] Dan Archdeacon, J Ellis-Monaghan, D Fisher, Dalibor Froncek, Peter Che Bor Lam, S Seager, Bing Wei, and Raphael Yuster. Some remarks on domination. *Journal of Graph Theory*, 46(3):207–210, 2004.

[6] Robert C Brigham, Phyllis Z Chinn, and Ronald D Dutton. Vertex domination-critical graphs. *Networks*, 18(3):173–179, 1988.

[7] Csilla Bujtás. Domination number of graphs with minimum degree five. *arXiv preprint arXiv:1912.13499*, 2019.

[8] Vašek Chvátal and Colin McDiarmid. Small transversals in hypergraphs. *Combinatorica*, 12:19–26, 1992.

[9] Ernest J Cockayne, RM Dawes, and Stephen T Hedetniemi. Total domination in graphs. *Networks*, 10(3):211–219, 1980.

[10] Alex Eustis, Michael A Henning, and Anders Yeo. Independence in 5-uniform hypergraphs. *Discrete Mathematics*, 339(2):1004–1027, 2016.

[11] David C Fisher, Patricia A McKenna, and Elizabeth D Boyer. Hamiltonicity, diameter, domination, packing, and biclique partitions of mycielski's graphs. *Discrete Applied Mathematics*, 84(1-3):93–105, 1998.

[12] Solomon W Golomb and Lloyd R Welch. Perfect codes in the lee metric and the packing of polyominoes. *SIAM Journal on Applied Mathematics*, 18(2):302–317, 1970.

[13] Masahisa Goto and Koji M Kobayashi. Connected domination in grid graphs. *arXiv preprint arXiv:2109.14108*, 2021.

[14] Sylvain Gravier and Abdelkader Khelladi. On the domination number of cross products of graphs. *Discrete mathematics*, 145(1-3):273–277, 1995.

[15] Jonathan L Gross and Thomas W Tucker. *Topological graph theory*. Courier Corporation, 2001.

[16] Teresa W Haynes, Stephen Hedetniemi, and Peter Slater. *Fundamentals of domination in graphs*. CRC press, 2013.

[17] Teresa W Haynes, Stephen T Hedetniemi, and Michael A Henning. *Topics in domination in graphs*, volume 64. Springer, 2020.

[18] Teresa W Haynes, Stephen T Hedetniemi, and Michael A Henning. *Domination in graphs: Core concepts*. Springer, 2023.

[19] Seiya Negami. The spherical genus and virtually planar graphs. *Discrete mathematics*, 70(2):159–168, 1988.

[20] Seiya Negami. Another approach to planar cover conjecture focusing on rotation systems. *Journal of the Mathematical Society of Japan*, 76(3):975–996, 2024.

[21] Bruce Reed. Paths, stars and the number three. *Combinatorics, Probability and Computing*, 5(3):277–295, 1996.

[22] David P Sumner. Critical concepts in domination. In *Annals of Discrete Mathematics*, volume 48, pages 33–46. Elsevier, 1991.

[23] Zsolt Tuza. Covering all cliques of a graph. *Discrete Mathematics*, 86(1-3):117–126, 1990.

[24] V.V. Vazirani. *Approximation Algorithms*. Springer Berlin Heidelberg, (2002).

[25] Vadim G Vizing. Some unsolved problems in graph theory. *Russian Mathematical Surveys*, 23(6):125–141, 1968.

[26] HB Walikar, BD Acharya, and E Sampathkumar. *Recent developments in the theory of domination in graphs and its applications*. 1979.

[27] Maciej Zwierzchowski. Domination parameters of a graph with added vertex. *Opuscula Mathematica*, 24(2):231–234, 2004.